

Problem

A remote employee needs to access the company private network from remote location securely without exposing the company internal network to internet. No other traffic should have access to company internal subnet.

Solution

VPC & Networking Setup

- Created VPC with IP block 10.0.0.0/16 for sufficient address range
- Built public subnet with EC2 VPN Gateway (internet-facing) + IGW route for connectivity
- Built private subnet for company resources/services, accesible only via VPN tunnel
- Added VPN overlay network route to private subnet route table (prevents packet drops)

Security Configurations

- Configured security groups for allowed traffic rules
- Applied NACLs at subnet level for granular control

Wireguard Implementation

- Launched EC2 instance in public subnet as WireGuard VPN sever (diagram gateway)
- Generated server/client public/private key pairs using wg genkey
- Created `/etc/wireguard/wg0.conf` with server private key, VPN subnet, UDP port 51820
- Enabled IP forwarding, started with wg-quick up wg0,
- Built client configs: client private key + server public key/EC2 IP + `AllowedIPs=10.0.0.0/16` for VPC access

Wireguard Configurations

Server Configurations

```
[Interface]
PrivateKey = <private-key>
Address = 10.200.0.1/24
ListenPort = 51820

[Peer]
PublicKey = 0zFzXif1Zu66gaKXVTOU4pKKGHh9aoxGKbMce5WafmE=
AllowedIPs = 10.200.0.2/32
```

Client Configurations

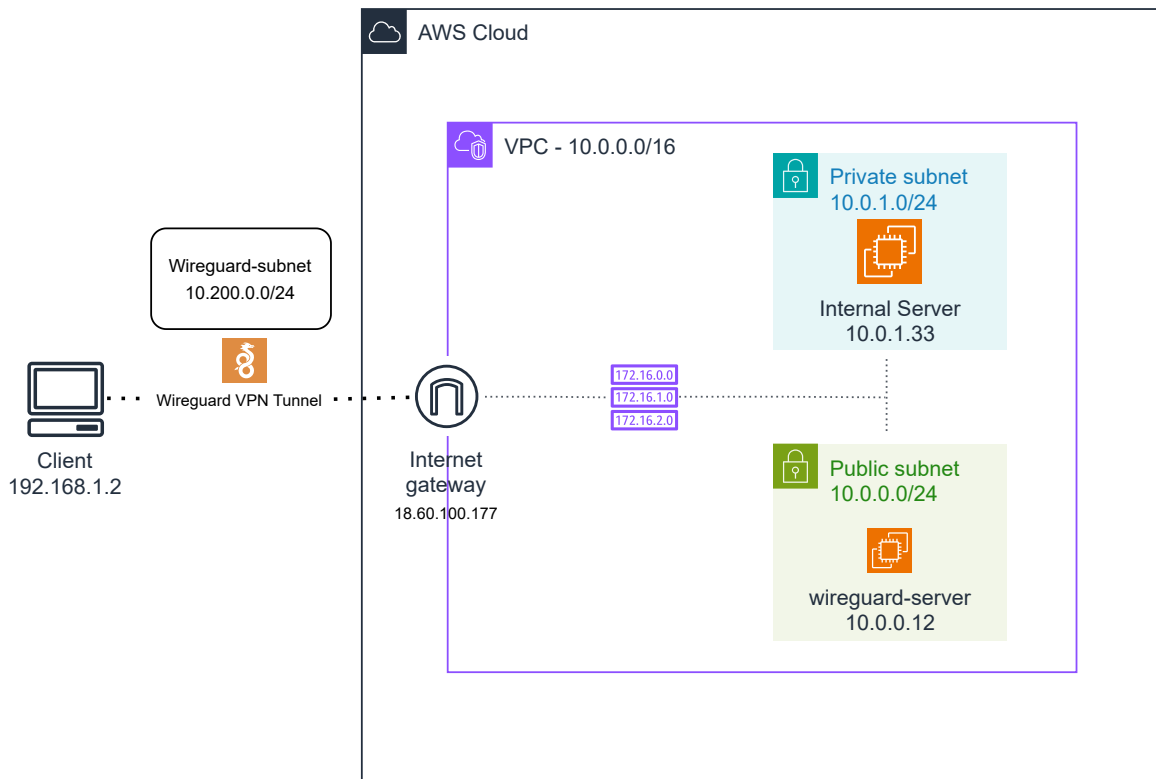
```
[Interface]

PrivateKey = <private-key>
Address = 10.200.0.2/24

[Peer]

PublicKey = HFBtRCBepMCW++VCHoHu9+IU2x1jyuwiP2v06GFn4QQ=
AllowedIPs = 10.200.0.0/24, 10.0.0.0/24
Endpoint = 18.60.100.177:51820
```

Diagram



Testing & Validation

Verifying wireguard handshake

```
saad@thinkpadT14:~$ sudo wg
interface: wg0
  public key: 0zFzXif1Zu66gaKXVTOU4pKKGHh9aoxGKbMce5WafmE=
  private key: (hidden)
  listening port: 54965
```

```
peer: HFBtRCBepMCW++VCHoHu9+IU2x1jyuwiP2v06GFn4QQ=
  endpoint: 18.60.100.177:51820
  allowed ips: 10.200.0.0/24, 10.0.1.0/24
  latest handshake: 2 minutes, 28 seconds ago
  transfer: 1.43 KiB received, 1.63 KiB sent
saad@thinkpadT14:~$
```

- ✓ Handshake successful - authentication and encryption working
- ✓ Data transfer confirmed - tunnel is active

Internal server access testing

pinging 10.0.1.33 (internal server)

```
saad@thinkpadT14:~$ ping 10.0.1.33
PING 10.0.1.33 (10.0.1.33) 56(84) bytes of data.
64 bytes from 10.0.1.33: icmp_seq=1 ttl=63 time=49.5 ms
64 bytes from 10.0.1.33: icmp_seq=2 ttl=63 time=23.9 ms
64 bytes from 10.0.1.33: icmp_seq=3 ttl=63 time=45.3 ms
^C
--- 10.0.1.33 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 23.945/39.592/49.516/11.196 ms
saad@thinkpadT14:~$
```

SSH connection test

SSH connection to internal server

```
saad@thinkpadT14:~/wireguardKeys$ sudo ssh ubuntu@10.0.1.33 -i vpnGW-ssh-key-pair.pem
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)
```

- * Documentation: <https://help.ubuntu.com>
- * Management: <https://landscape.canonical.com>
- * Support: <https://ubuntu.com/pro>

System information as of Wed Jan 21 03:55:39 UTC 2026

System load:	0.0	Temperature:	-273.1 C
Usage of /:	26.5% of 6.71GB	Processes:	113
Memory usage:	23%	Users logged in:	0
Swap usage:	0%	IPv4 address for ens5:	10.0.1.33

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

```
Last login: Sat Jan 17 11:56:16 2026 from 10.200.0.2
ubuntu@ip-10-0-1-33:~$ ls
internal_server
ubuntu@ip-10-0-1-33:~$
```

Verifying VPN routes

```
saad@thinkpadT14:~$ ip route | grep wg0
10.0.0.0/24 dev wg0 scope link
10.0.1.0/24 dev wg0 scope link
10.200.0.0/24 dev wg0 proto kernel scope link src 10.200.0.2
saad@thinkpadT14:~$
```

- ✓ Traffic to VPN subnet (10.200.0.0/24) through tunnel
- ✓ Traffic to internal server (10.0.1.0/24) through tunnel